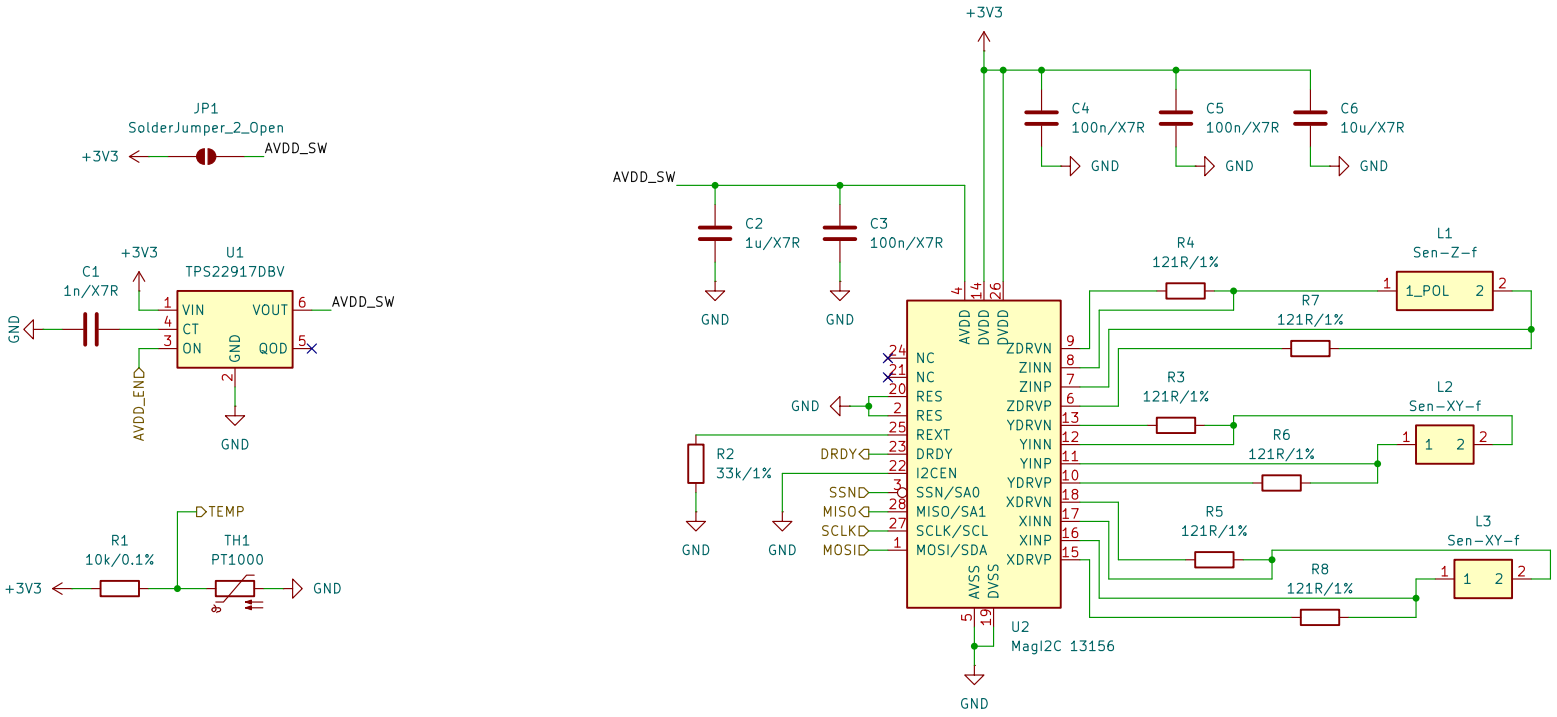


Sheet: /
File: attinav.kicad_sch

Title:

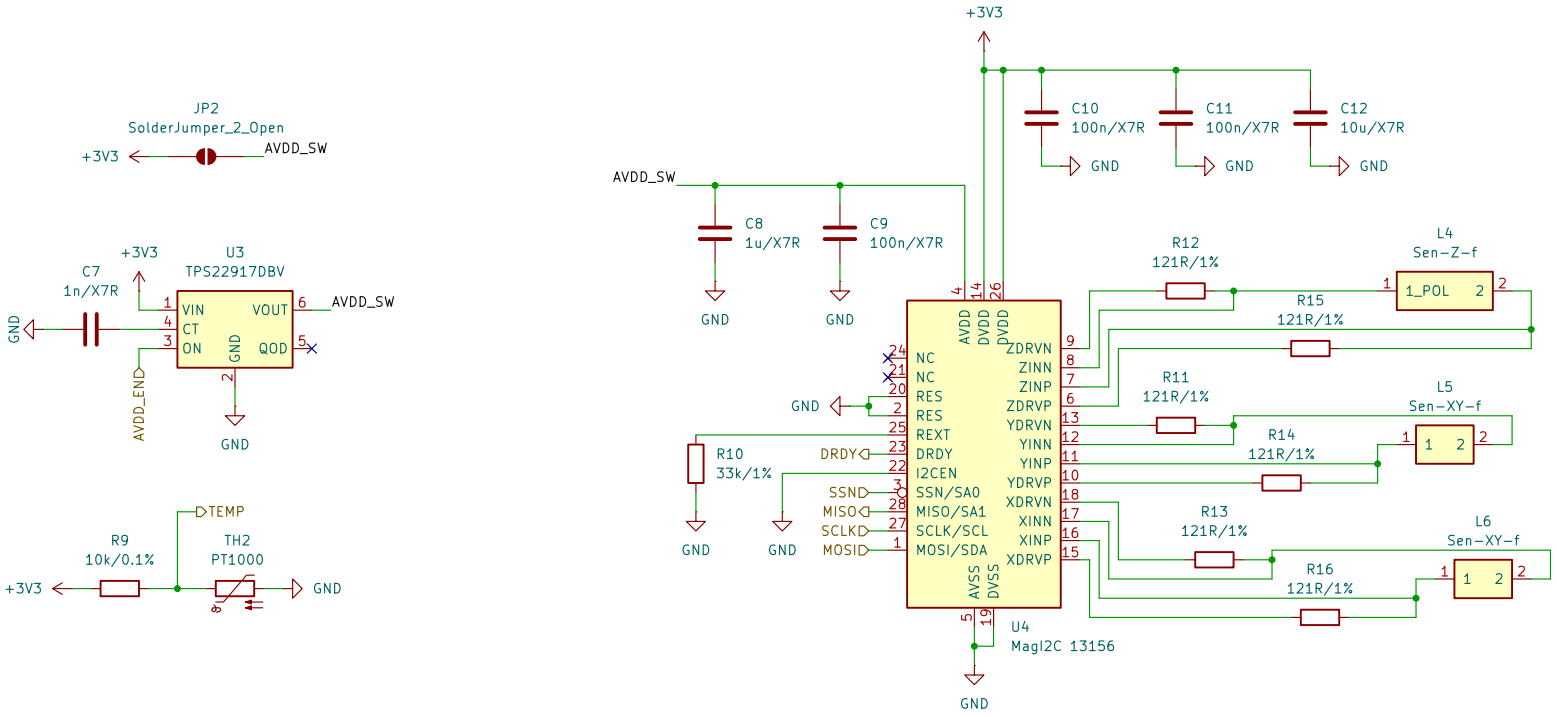
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 1/8



RM3100 triad. Keep caps/resistors away from coils; decouple at MagI2C.
121R in series with each DRV pin (PNI manual R07).

AVDD switchable for quiet-window power gating; close JP1 to bypass U1.

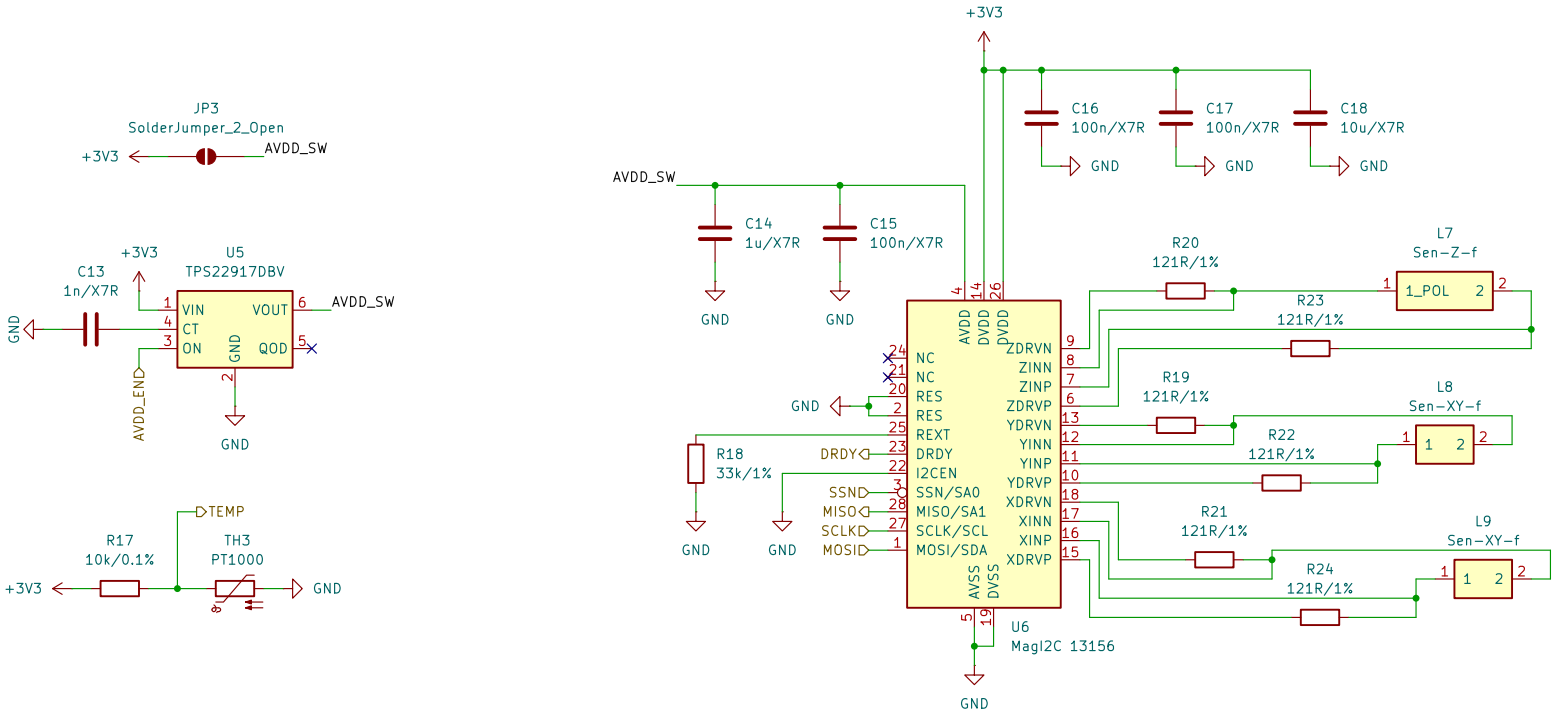
Sheet: /MAG1/		
File: mag.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 2/8



RM3100 triad. Keep caps/resistors away from coils; decouple at MagI2C.
121R in series with each DRV pin (PNI manual R07).

AVDD switchable for quiet-window power gating; close JP1 to bypass U1.

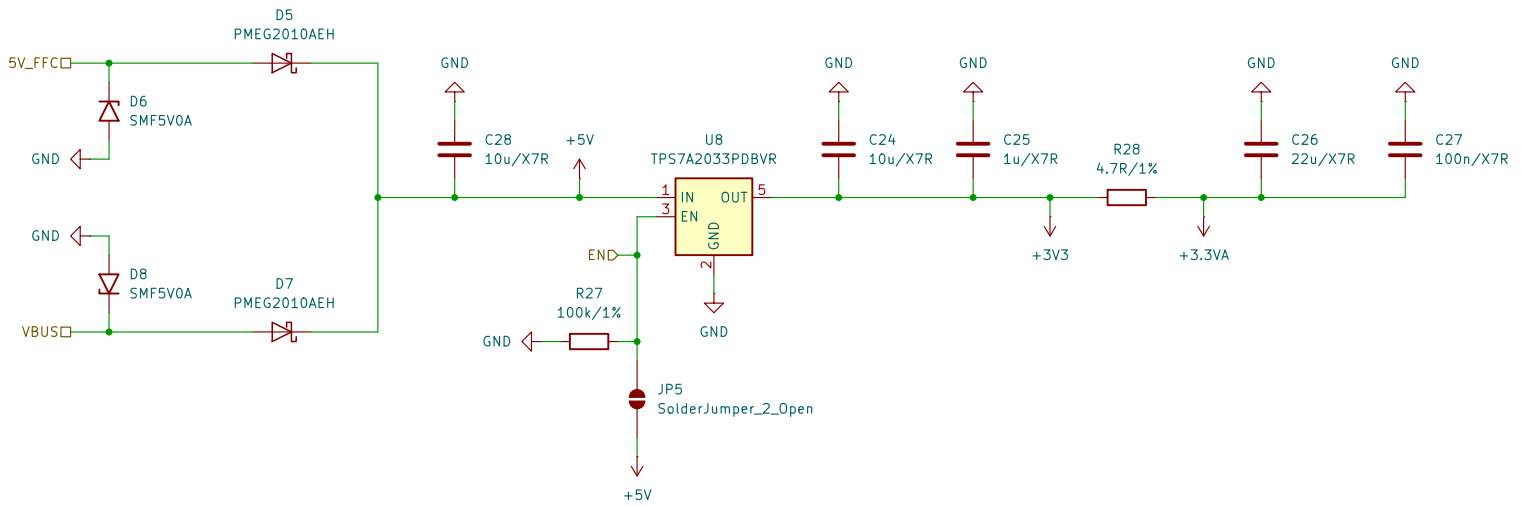
Sheet: /MAG2/		
File: mag.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 3/8

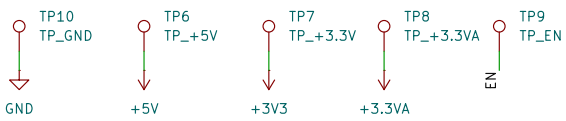


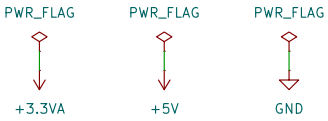
RM3100 triad. Keep caps/resistors away from coils; decouple at MagI2C.
121R in series with each DRV pin (PNI manual R07).

AVDD switchable for quiet-window power gating; close JP1 to bypass U1.

Sheet: /MAG3/		
File: mag.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 4/8



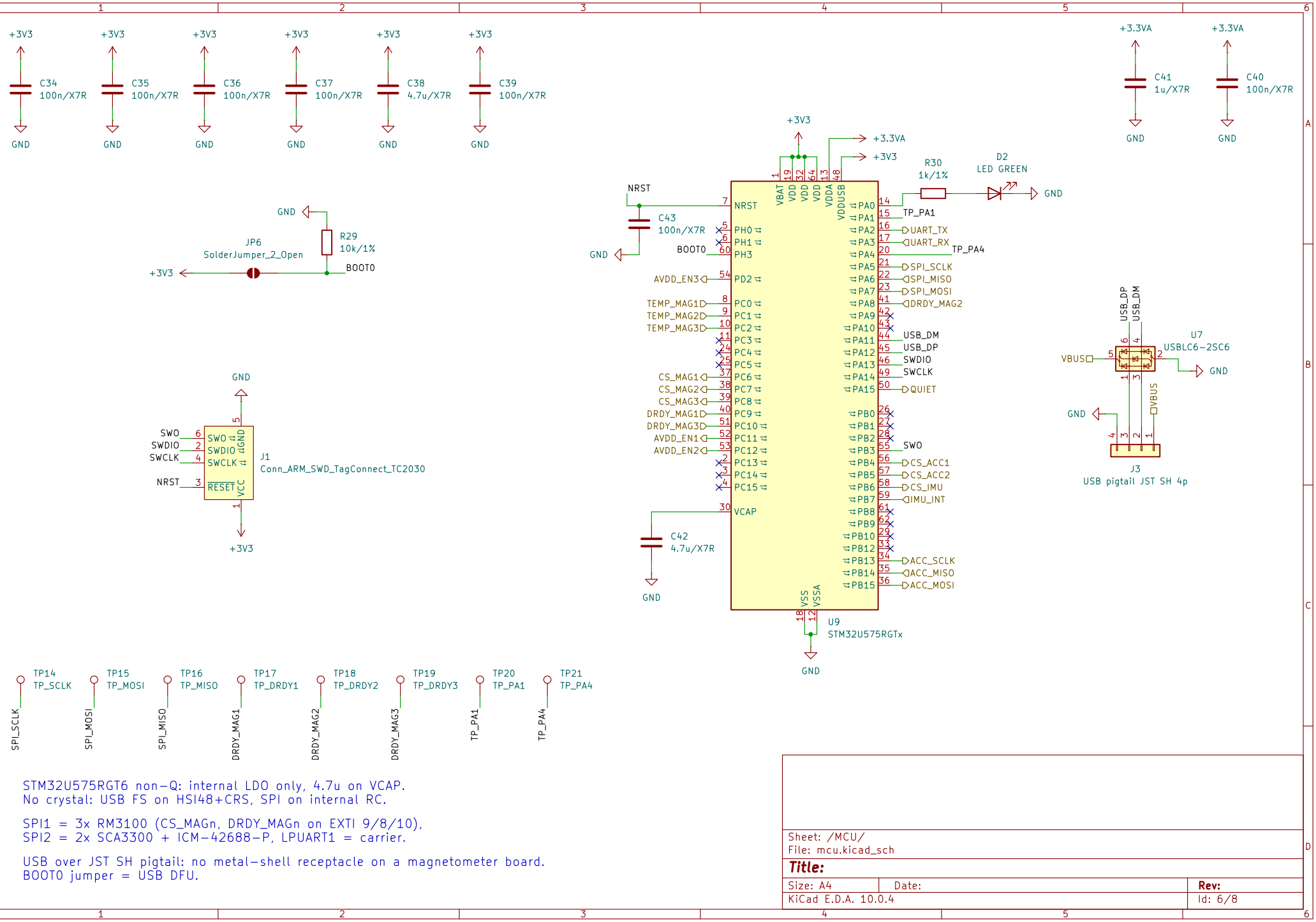


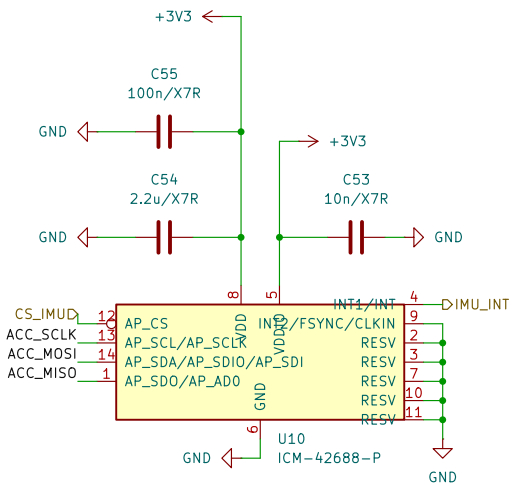


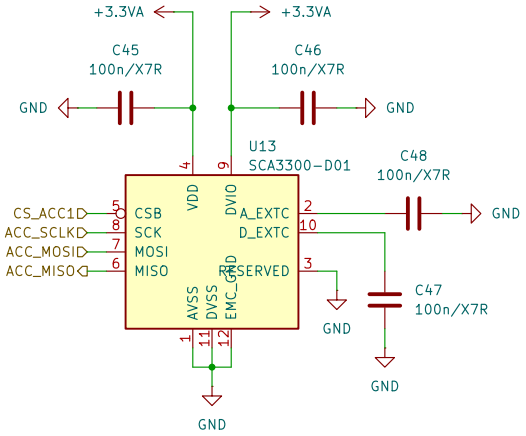
Power: 5 V from FFC or USB VBUS, Schottky OR-ed, TVS on each input.
TPS7A2033 LDO 300 mA, 7 uVrms, high PSRR.

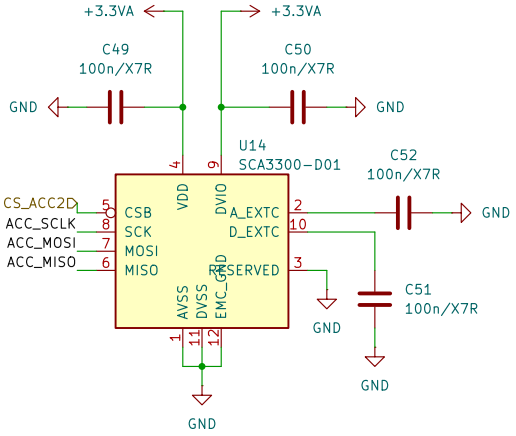
EN: 100k pulldown (default off); JP5V->EN for bench enable. +3.3VA: RC, no ferrite (non-magnetic).

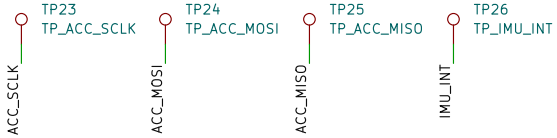
Sheet: /POWER/		
File: power.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 5/8


+3V3
C34
100n/X7R
GND
C35
100n/X7R
C36
100n/X7R
C37
100n/X7R
C38
4.7u/X7R
C39
100n/X7R
+3.3VA
C41
1u/X7R
C40
100n/X7R
JP6
SolderJumper_2_Open
R29
10k/1%
BOOT0
R30
1k/1%
D2
LED GREEN
NRST
C43
100n/X7R
BOOT0
AVDD_EN3
TEMP_MAG1
TEMP_MAG2
TEMP_MAG3
CS_MAG1
CS_MAG2
CS_MAG3
DRDY_MAG1
DRDY_MAG3
AVDD_EN1
AVDD_EN2
C42
4.7u/X7R
VCAP
VBAT
VDD
VDDA
VDDUSB
VSS
VSSA
U9
STM32U575RGTx
TP_PA1
UART_TX
UART_RX
TP_PA4
SPI_SCLK
SPI_MISO
SPI_MOSI
DRDY_MAG2
USB_DM
USB_DP
SWDIO
SWCLK
QUIET
SWO
CS_ACC1
CS_ACC2
CS_IMU
IMU_INT
ACC_SCLK
ACC_MISO
ACC_MOSI
SWO
SWDIO
SWCLK
NRST
J1
Conn_ARM_SWD_TagConnect_TC2030
U7
USBLC6-2SC6
VBUS
J3
USB pigtail JST SH 4p
TP14
TP_SCLK
TP15
TP_MOSI
TP16
TP_MISO
TP17
TP_DRDY1
TP18
TP_DRDY2
TP19
TP_DRDY3
TP20
TP_PA1
TP21
TP_PA4
STM32U575RGT6 non-Q: internal LDO only, 4.7u on VCAP.
No crystal: USB FS on HSI48+CRS, SPI on internal RC.
SPI1 = 3x RM3100 (CS_MAGn, DRDY_MAGn on EXTI 9/8/10),
SPI2 = 2x SCA3300 + ICM-42688-P, LPUART1 = carrier.
USB over JST SH pigtail: no metal-shell receptacle on a magnetometer board.
BOOT0 jumper = USB DFU.
Sheet: /MCU/
File: mcu.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.4
Id: 6/8







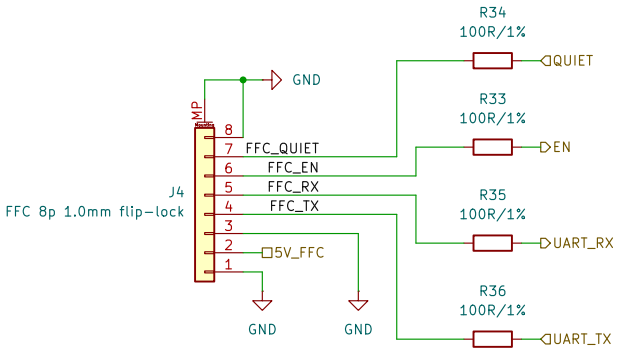


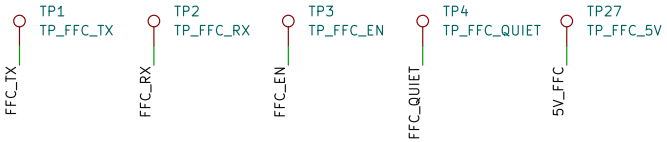
2x SCA3300-D01 Mode 4 on SPI2. VDD and DVIO both on +3.3VA: DVIO must never exceed VDD (Murata DS).

LAYOUT: rotate ACC2 90 deg from ACC1 about Z; sight axis on X/Y, never Z (Z offset tempco 15 mg vs 10 mg).

ICM-42688-P = motion/validity flag only, on +3V3; FSYNC and RESV to GND per TDK DS.

Sheet: /ACCEL/ File: accel.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 7/8





FFC to carrier, 1.0 mm pitch, 8 pos: GND 5V GND TX RX EN QUIET GND (TX/RX named from this module's side).

EN high = on (100k pull-down on POWER).
QUIET = open-drain from MCU, pull-up on the carrier. 100R series on every signal.

Prefer an all-plastic-body FFC part if available;
metal hold-downs are over 55 mm from the coils and calibrated out.

Sheet: /INTERFACE/ File: interface.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 8/8